JMO – 2004

Max Mark – 100 Time – 3 Hrs

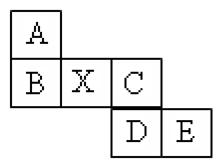
Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all questions.

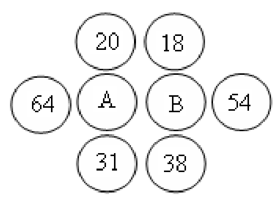
1.
   (a) If the following numbers are arranged in the ascending order. Then which number lies in the middle? $\frac{1}{3}$, $\frac{3}{10}$, 31%, 0.03, 0.303

   (b) If measure of smallest angle of a triangle is 20^0, then what is the at most measure of largest angle of that triangle? (i) 80^0 (ii) 90^0 (iii) 140^0 (iv) 1590^0 (v) 160^0

2.
   (a) If the given below paper folded and make a cube, then what is the opposite to X?

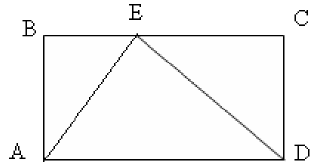


   (b) Ali (A) and Baba (B) rounded by 6 thieves. Ages of thieves are written inside the circles. If age of A is same as average of nearest 4 peoples and age of B is also average of nearest 4 peoples. Then what is the age of Ali?

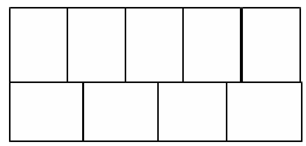


3. In a box there are some cubes, cones and spheres. The weight of 2 cones and one sphere is same as a cube. Similarly weight of one sphere and one cube is same as three cones. Then weight of how many spheres is same as one cone?
4. A runs faster than B. In a running competition D defeated C and E never defeat E. One day a competition held between these five persons. Then which one is true? Where (ABCDE means A is first, B is second, C is third, D is fourth and E is fifth).
   (i) ABCDE (ii) BEDAC (iii) ABCED (iv) ADBCE (v) ADCEB.
5. If sum of two prime numbers is 999, then what is their product?
6. In Bijay's birthday he is 14 years old while his father is 41 years old. He realizes that when he reverses the digit of his age he got his father's age. After this in which birthday the same will happen?
7. The cost of an article decreases from Rs 25 to Rs. 16 after giving two consecutive equal % of discount. Find % of discount given in each time.
8. Fill in the blanks such that equation is true. _ _ % of _ _ _ = 400.
9. .
10. If $n! = 1\times 2\times 3\times 4\times ....\times n$. Example $12! = 1\times 2\times 3\times 4\times ....\times 12$. Find the value of n for which $n! = (3!)(5!)(7!)$.
11. In mathematics department out of 100 students 99% are girls. Some students are staying in campus. Out of all the campus students 98% are girls. Then how many students staying outside the campus?

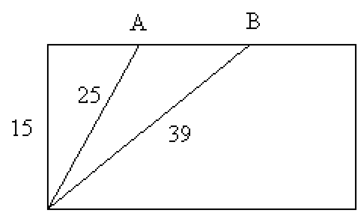
12. In the figure AE = 3, DE = 4, AD = 5. Then find the area of rectangle ABCD.



13. Find the number of perfect square in between 20 and 9999.
14. 9 squares are like figure below such that it makes a rectangle whole area is 180 sq units. Then find the perimeter of the rectangle.



15. After simplification $10^{101} - 1$, we got a number. Find the sum of all the digits of that number.
16. In a clock there is no minute hand but still its hour hand shows right time. If hour hand pointed towards 22 minutes, then what time shows by the clock?
17. In two boxes equal numbers of balls are there. Balls are of either red or white in colour. In first box ratio of red and white balls is 7:1 and in second box ratio of red and white balls is 9:1. If in total 90 balls are white, then how many red balls are there in second box?
18. n = * * 1 9 * * * * * * * 9 7 * * * is a 16 digit number where * consists of digits. Sum of any four consecutive digits of the number is 24. Then find the sum of all the 7 digits between two 9's.
19. In the figure A and B are point lies on one side of a rectangle and joint by vertical point of the rectangle such that length of two lines are 25 and 39. Small side of rectangle is 15 then find length of AB.



20. Find the missing digits denoted by * in the following addition.

$$\begin{array}{r} 8\ 2\ * \\ 1\ *\ 9 \\ 1*\ 6\ 4 \\ \hline 1\ *\ *\ 9 \end{array}$$

21. You start counting from 1 to 1000 in your finger. Starting from 1 thumb, 2 second finger, 3 middle finger, 4 ring finger, 5 small finger, 6 ring finger, 7 middle finger, 8 second finger, 9 thumb, 10 second finger, 11 middle finger, so on. Then which finger is 1000th finger?

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